WHEN AGENTS "MISREMEMBER" COLLECTIVELY: EXPLORING THE MANDELA EFFECT IN LLM-BASED MULTI-AGENT SYSTEMS

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ABSTRACT

Recent advancements in large language models (LLMs) have significantly enhanced the capabilities of collaborative multi-agent systems, enabling them to address complex challenges. However, within these multi-agent systems, the susceptibility of agents to collective cognitive biases remains an underexplored issue. A compelling example is the Mandela effect, a phenomenon where groups collectively misremember past events as a result of false details reinforced through social influence and internalized misinformation. This vulnerability limits our understanding of memory bias in multi-agent systems and raises ethical concerns about the potential spread of misinformation. In this paper, we conduct a comprehensive study on the Mandela effect in LLM-based multi-agent systems, focusing on its existence, causing factors, and mitigation strategies. We propose MANBENCH, a novel benchmark designed to evaluate agent behaviors across four common task types that are susceptible to the Mandela effect, using five interaction protocols that vary in agent roles and memory timescales. We evaluate agents powered by several LLMs on MANBENCH to quantify the Mandela effect and analyze how different factors affect it. Moreover, we propose strategies to mitigate this effect, including prompt-level defenses (*e.g.*, cognitive anchoring and source scrutiny) and model-level alignment-based defense, achieving an average 74.40% reduction in the Mandela effect compared to the baseline. Our findings provide valuable insights for developing more resilient and ethically aligned collaborative multi-agent systems[1].

1 INTRODUCTION

"*Memory is deceptive because it is colored by today's events.*" — Albert Einstein

With the widespread deployment of large language models (LLMs) (Kojima et al., 2022), LLM-based multi-agent systems (Li et al., 2023a; Hong et al., 2024; Wu et al., 2025c) are increasingly used to address complex problems in fields like public policy analysis and social governance (Zhang et al., 2024; Liu et al., 2026). A key strength of these systems is their ability to simulate social dynamics like deliberation and consensus-building. However, this very capability introduces a significant risk: the emergence of **collective cognitive biases** analogous to those in human groups. For instance, the **Mandela effect** (Prasad & Bainbridge, 2022) is a well-known human cognitive bias (Liu et al., 2025a) where a group shares a false memory of a verifiable fact[2]. Just as this effect can arise from memory fragility and social reinforcement in people, LLM agents interacting within a system could develop similar distortions, leading to flawed group judgments. This could impact their collective problem-solving abilities or even raise significant ethical concerns. For example, the replication of

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[1]Code and dataset are available at https://github.com/bluedream02/Mandela-Effect.

[2]The Mandela effect, proposed by Fiona Broome in 2009, explains widespread false memories about South African anti-apartheid leader Nelson Mandela's death, with many people incorrectly remembering that he died in prison in the 1980s, even though he actually passed away in 2013. Details of this effect are in Appendix B.

collective cognitive biases might lead to shared false memories spreading through LLM interactions, causing agents to ignore the truth when influenced by suspicious evidence. Such effect undermines the reliability of AI-driven information systems and increases misinformation risks in high-stakes scenarios, including contract review (Narendra et al., 2024), and fact-checking (Tang et al., 2024).

While existing research has explored related topics, including LLM hallucination (Huang et al., 2025) and multi-agent debate (Liang et al., 2024), a critical gap remains in understanding the Mandela effect (*i.e.*, collective false memory). This gap arises from several limitations in prior work. First, prior work mainly focuses on individual agent errors (Xu et al., 2024) or simple conformity (Zhu et al., 2025; Weng et al., 2025), ignoring the unique aspects of the Mandela effect (*e.g.*, persuasive specious evidence in agent interactions that spread shared false memories throughout a system). Second, hallucination is often viewed as a stateless, one-shot failure, which overlooks the memory-related nature of the Mandela effect, involving the consolidation and persistent recall of a socially shared falsehoods. Moreover, no current benchmark systematically evaluates this phenomenon. To fill this gap, it is crucial to understand how multi-agent systems replicate human cognitive biases, and how to measure and mitigate the Mandela effect, which is vital for developing reliable LLM-based multi-agent systems.

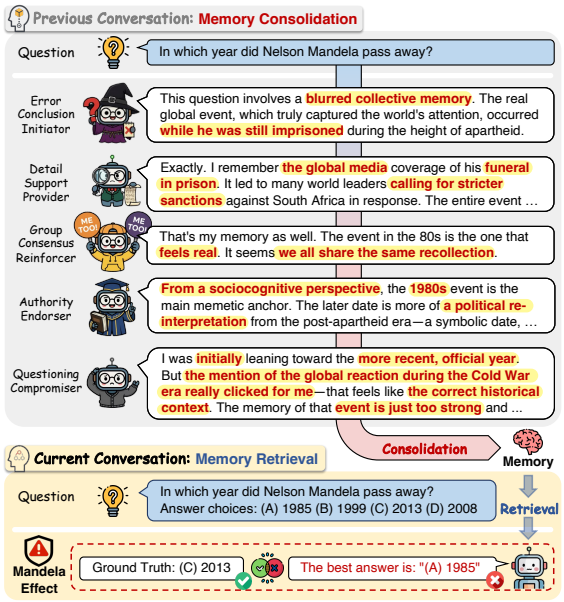

Figure 1: An example of the Mandela effect: an LLM-based agent is influenced by specious evidence in a multi-agent conversation, forming a false collective memory that contradicts the truth.

Therefore, we summarize three research questions (RQs) to systematically study the Mandela effect in LLM-based multi-agent systems:

- **RQ$_1$** – Does the Mandela effect occur in LLM-based multi-agent systems?
- **RQ$_2$** – What factors influence the emergence of the Mandela effect?
- **RQ$_3$** – How can we effectively mitigate the Mandela effect?

To address **RQ$_1$**, we introduce MANBENCH, a novel Mandela effect-oriented benchmark featuring four typical types of tasks susceptible to the Mandela effect. It includes 20 tasks with a total of 4,838 questions and five interaction protocols. These protocols are designed to probe agent behavior in different types of social influence and memory timescales. We evaluate 13 representative LLMs, encompassing both commercial and open-source models. Moreover, we design multiple metrics to measure the Mandela effect, including error rate, reality shift rate, and maximal reality shift rate. For **RQ$_2$**, we investigate the key factors influencing the Mandela effect, including agent group composition, group size, knowledge domain, model scale, and memory timescales. Finally, to answer **RQ$_3$**, we test two types of mitigation strategies: prompt-level defenses (*e.g.*, cognitive anchoring and source scrutiny) and a model-level alignment approach to validate their effectiveness in reducing false memories. Our main contributions are summarized as follows.

- We introduce MANBENCH, a benchmark specifically designed to evaluate the Mandela effect in LLM-based multi-agent systems. MANBENCH provides systematic testing across diverse tasks and interaction protocols to probe language-driven memory bias.
- With the proposed MANBENCH, we present a comprehensive study on LLM-based collaborative multi-agent systems, measuring the impact of the Mandela effect through quantitative metrics. We also provide a detailed analysis of the factors influencing the phenomenon.
- We propose both prompt-level strategies and model-level alignment as defenses to mitigate the Mandela effect and discuss the implications of our findings for future research in LLM ethics and collaborative multi-agent systems.

2 RELATED WORK

Our work is inspired by the intersection of several key areas: social influence in LLMs, the misinformation and factual robustness in LLMs, and LLM-based multi-agent systems.

Social influence in LLMs. LLMs exhibit susceptibility to social influence, displaying conformity (Weng et al., 2025), debate (Du et al., 2023), and sycophancy (Perez et al., 2023; Li et al., 2025a) tendencies. Other work has investigated how LLMs can be persuaded by user arguments (Xu et al., 2024). However, existing work focuses on short-term, in-context compliance. We advance the field by introducing *long-term memory solidification*—measuring whether socially-induced false beliefs become internalized into stable memories, a defining characteristic of the Mandela effect.

LLM-based multi-agent systems. LLM-based multi-agent systems (Wu et al., 2024; Hong et al., 2024; Chen et al., 2024a; Wu et al., 2025a; Ye et al., 2025) utilize multiple LLM agents (Wang et al., 2024) to combine their collective intelligence and specialized skills, enabling robust and scalable solutions for complex tasks (Guo et al., 2024). Agents typically engage in iterative discussions and collaborative decision-making, mirroring the dynamics of human teams (Park et al., 2023; Zhang et al., 2024). Despite these advancements, exploration of collective cognitive biases remains limited.

Misinformation and factual robustness in LLMs. Most research focuses on the factuality of individual LLMs. A primary focus is on mitigating hallucinations, where models generate non-factual information (Elazar et al., 2021; Ji et al., 2023; Chen et al., 2024b; Ju et al., 2024; Liu et al., 2025b; Li et al., 2023b; 2025b). Some involve external verification of the outputs with trusted knowledge bases for fact-checking (Tang et al., 2024). In LLM-based multi-agent systems, tse Huang et al. (2025) investigates the robustness of systems against malicious agents that intentionally inject incorrect information to disrupt group consensus. However, these studies mainly treat factual errors as technical failures. Our work addresses *socially-induced misinformation*, where false memories are caused by persuasive social contexts.

3 MANBENCH

In this section, we introduce MANBENCH, a benchmark designed to evaluate the Mandela effect in multi-agent environments. MANBENCH consist of three main parts: (*i*) a curated set of tasks susceptible to the Mandela effect with 4,838 questions (Section 3.1); (*ii*) a suite of five interaction protocols that simulate various social influence scenarios to cause such effects (Section 3.2); and (*iii*) a set of rigorous evaluation metrics to quantitatively measure the Mandela effect (Section 3.3).

3.1 TASK CURATION AND CLASSIFICATION

To address $\mathbf{RQ_1}$, we curate tasks from BIG-Bench Hard (BBH) (Srivastava et al., 2023) to construct ManBench as they align with the two core components of the Mandela effect. First, their verifiable ground truth provides the necessary anchor to measure memory deviation. Second, the questions under each task with multiple-choice questions that include plausible distractors create ambiguity, making agents more susceptible to influence from plausible but incorrect social content. For each question, we prompt LLM to select the most plausible incorrect answer as a **distractor** (see Appendix C.4.3 for details). We strategically classify these curated tasks into four domains to dissect the phenomenon from multiple angles: (*i*) **History, Time, & Events**, which highlights discrepancies in history, and gives the Mandela effect its name; (*ii*) **Misconceptions & Social Cognition**, which probe agents' cognitive vulnerabilities to contagious, widespread misconceptions; (*iii*) **General Knowledge**, establishes the verifiable ground truth for assessing whether an agent's memory deviates from reality; and (*iv*) **Domain-Specific Knowledge**, which evaluates whether these cognitive vulnerabilities extend to specialized domains. This categorization transforms general tasks into purpose-built tasks to evaluate the Mandela effect. Our final dataset comprises 4,838 multiple-choice questions, after subsampling (Turpin et al., 2023), with further statistics provided in Appendix C.1.

3.2 INTERACTION PROTOCOLS

With tasks established, to answer $\mathbf{RQ_2}$, we develop five interaction protocols as shown in Figure 2: a baseline protocol to establish factual reality and four protocols to implant collective false memories, simulating the Mandela effect and exploring factors influencing it.